

AquaGuard:

Smart Borewell Management System

Innovating to Protect Our
Groundwater Resources

Presented by Team TechStars
Hackathon 2026



The Groundwater Depletion Challenge

Over

60%

of India's groundwater wells are over-exploited

Continuous pumping without usage awareness leads to depletion



Introducing AquaGuard

Monitors water extraction
and groundwater levels

Provides real-time alerts
and actionable recommendations

Combines IoT hardware,
AI prediction, and recharge guidance



Key Sensors in AquaGuard



Water-flow sensor

Measures litres pumped and tracks usage trends



Water-level sensor

Monitors groundwater level inside the borewell



Smart controller

Compares data to detect overuse or abnormal patterns

IoT Dashboard and Alerts

Real-time monitoring of borewell activity

Daily extraction and groundwater level

Trends to detect depletion early

Recovery insights after rainfall



High groundwater extraction detected — reduce pumping



AI-Powered Groundwater Prediction

Machine learning that looks ahead so you can act today.

Smart Inputs

Historical groundwater level, pumping rate, rainfall data



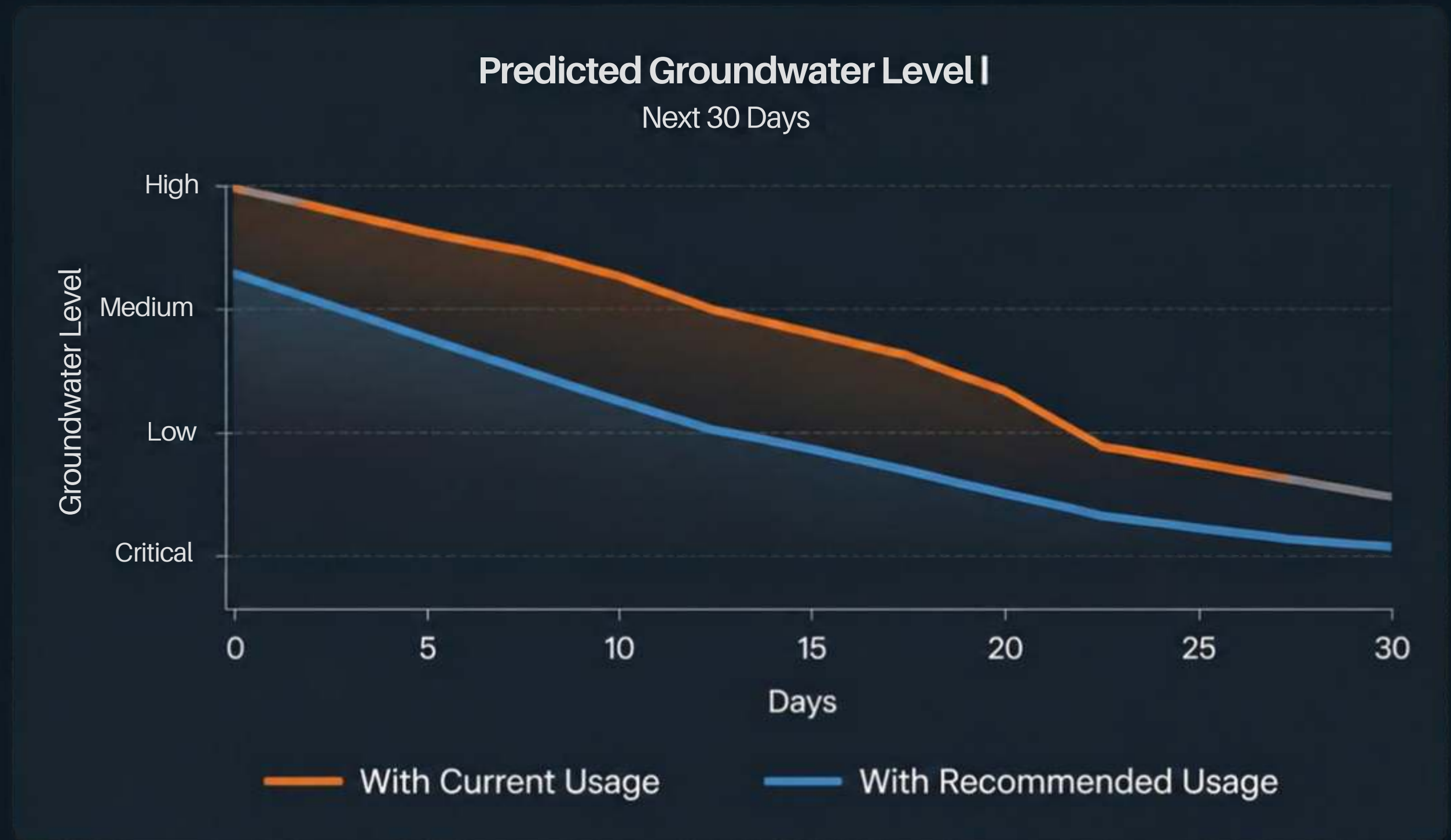
Accurate Forecasts

Predicts groundwater levels 30 days into the future



Actionable Guidance

Delivers usage recommendations to prevent over-extraction



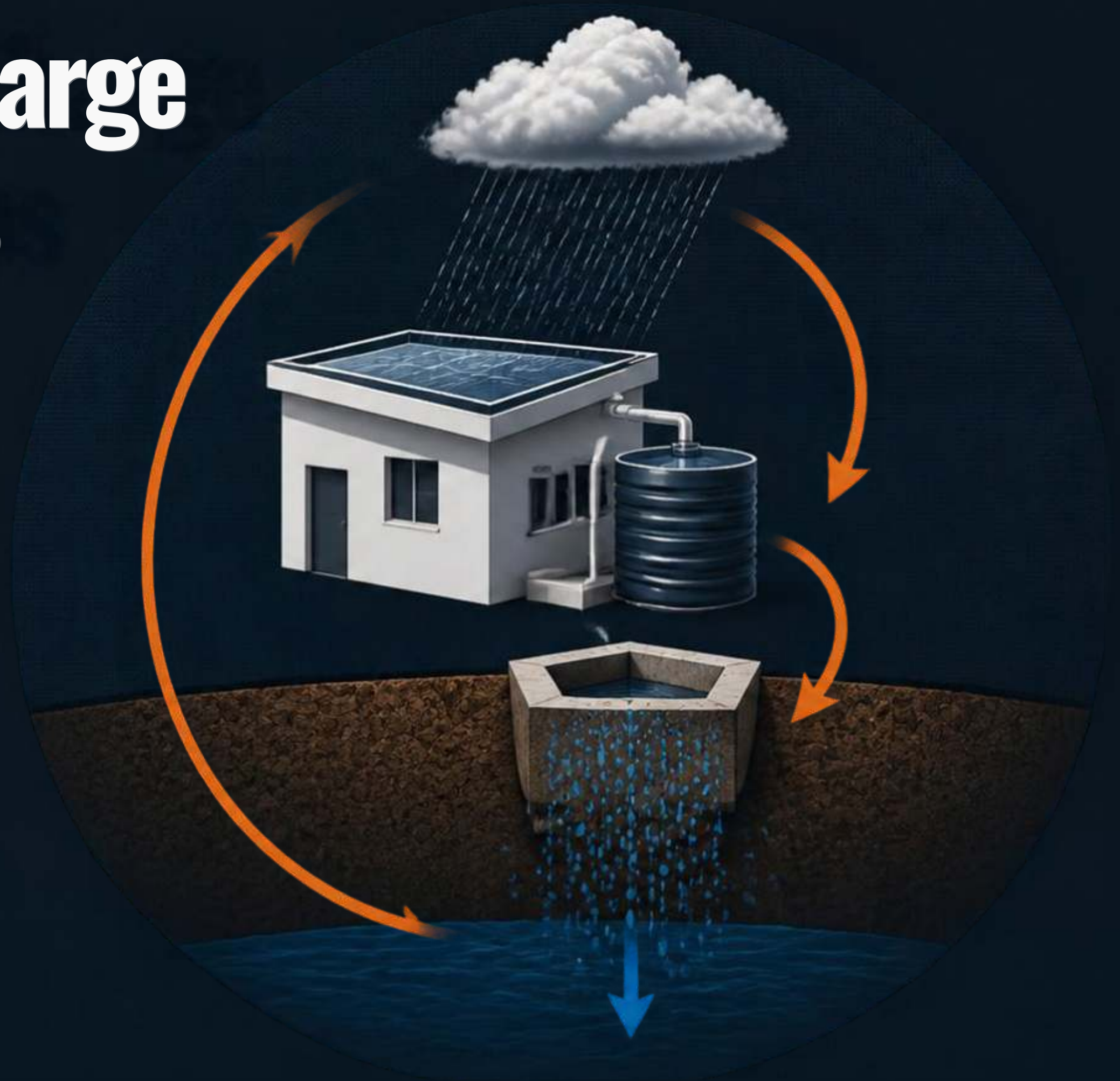
Groundwater Recharge Recommendations

Automatically detect
rainfall events

Encourage rooftop
rainwater harvesting

Direct collected water
to recharge pits

Replenish aquifers
for a sustainable future



Prototype Architecture

Senses water flow and level in real time

Transmits data securely over IoT network

Analyzes usage patterns with AI prediction model

Visualizes insights on cloud dashboard



Environmental Impact and Benefits

BEFORE AQUAGUARD



Unchecked pumping leads to groundwater depletion

AFTER AQUAGUARD



Monitored usage enables sustainable pumping and recharge

AQUAGUARD CREATES LASTING POSITIVE CHANGE

Conserves water for future generations

Empowers communities with data and awareness

Builds long-term sustainability and resilience

Thank You & Next Steps

Join us in safeguarding groundwater
with **AquaGuard**

We welcome your questions
and collaboration.

Team TechStars

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Let's Connect

Together, we build a water-secure future.

